



Advancing financial risk management: A transparent framework for effective fraud detection

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ABSTRACT

Robust financial fraud detection is crucial for protecting assets and maintaining financial system integrity. Traditional models lack flexibility, while machine learning models are often complex and difficult to interpret. We propose an XGB-GP framework that combines Extreme Gradient Boosting (XGB) and Genetic Programming (GP) to create interpretable models, enhancing fraud detection. Our framework highlights the effectiveness of the financial indicator “Total Liabilities/Operating Costs” and outperforms traditional and machine learning models in detecting fraud, as demonstrated through analysis of data from the CSMAR database of Chinese publicly listed companies.

1. Introduction

Financial fraud significantly undermines market stability by deceiving stakeholders and concealing actual financial risks through the manipulation of income, assets, or liabilities. In addition to causing substantial losses to investors and heightening the risk of future stock market crashes (Wen et al., 2023), financial fraud disrupts resource allocation and impedes the growth of the securities market (Amar et al., 2022). Consequently, financial fraud serves as a key catalyst for financial risk.

As fraud tactics become more sophisticated, involving coordinated manipulation of financial statements, traditional detection methods often fall short. Early studies primarily relied on statistical approaches for fraud detection. Beneish (1999) introduced the M-score model, a numerical indicator to assess the likelihood of financial fraud. With the rise of machine learning, advanced models such as Neural Networks, Decision Trees, and Support Vector Machines have substantially improved the accuracy of fraud detection. For instance, Achakzai and Juan (2022) employed machine learning meta-classifiers to detect financial frauds, achieving a significant improvement in prediction accuracy over traditional statistical methods. Recently, hybrid models have garnered increasing attention in financial fraud detection. For example, Duan et al. (2022) proposed an integrated machine learning framework to predict financial fraud, yielding promising detection results. Despite these advancements, however, most machine learning-based models suffer from a lack of interpretability, which remains a key limitation and hampers their practical application and broader adoption (Ding et al., 2023).

Feature selection remains a crucial aspect of fraud detection, with previous research emphasizing the significance of financial indicators in identifying fraudulent activities (Zhao et al., 2024). In 1966, Beaver conducted pioneering work, applying the predictive power of financial indicators in empirical studies to detect financial fraud (Beaver, 1996). As machine learning models continue to evolve in the field of fraud detection, increasing attention has been given to feature selection to enhance model performance. Hajek

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Table 1
Classification table of company violations in CSMAR database.

Code	Violation type	Code	Violation type
P2501	Fabricated Profits	P2509	Unauthorized Change of Fund Usage
P2502	Fabricated Assets	P2510	Misappropriation of Company Assets
P2503	False Records	P2511	Insider Trading
P2504	Delayed Disclosure	P2512	Illegal Stock Transactions
P2505	Major Omissions	P2513	Stock Price Manipulation
P2506	Inaccurate Disclosure	P2514	Illegal Guarantees
P2507	Fraudulent IPO	P2515	Improper General Accounting Practices
P2508	Illegal Investment	P2599	Others

and Henriques (2017) utilized feature selection techniques to preprocess financial data, resulting in a significant improvement in the performance of models. However, manual feature selection methods are often inefficient and prone to errors, underscoring the need for automated, robust, and reliable approaches to improve the accuracy and accessibility of detection systems.

Ribeiro et al. (2016) emphasized the importance of incorporating interpretability into financial fraud detection models, noting that the “black box” nature of many machine learning models creates challenges for regulatory compliance and trust. In financial fraud detection, transparency and interpretability are critical for fostering auditors’ confidence in the model’s predictions, particularly in high-risk scenarios. This study addresses these challenges by introducing the XGB-GP model, a hybrid approach that integrates XGBoost for feature selection with Genetic Programming to enhance interpretability. Unlike traditional “black box” models, XGB-GP produces interpretable equations, providing actionable insights that are accessible to both technical and non-technical users.

Albizri et al. (2019) critiqued the over-reliance on traditional financial ratios, advocating for the integration of advanced analytical techniques and non-financial indicators to strengthen fraud detection frameworks. Previous research indicates that traditional methods for selecting fraud detection features often rely on manual judgment and statistical approaches. While these methods can detect anomalies to some extent, they struggle to capture complex patterns and adapt to evolving fraud tactics. In this study, the Point-Biserial Correlation method is first employed to identify features with linear relationships to financial fraud, ensuring that each selected ratio is economically significant. The XGBoost model is then used to automatically select the most important features, utilizing the “gain” metric to identify features with complex nonlinear correlations to fraud. To further investigate the potential relationships between financial characteristics and fraudulent activities, this study integrates both traditional financial indicators and all generated financial ratios into the screening model for a comprehensive analysis. This approach facilitates the identification of novel key indicators for fraud detection, enabling real-time updates to the model and improving its ability to effectively capture fraudulent activities.

Through a joint analysis of financial and non-financial ratios using machine learning, the model identified key indicators of financial fraud, including five financial ratios such as Return on Equity (ROE), Current Ratio, and the newly introduced Total Liabilities to Operating Costs ratio. Notably, this study highlights the importance of the Total Liabilities to Operating Costs ratio, a feature often overlooked in previous research, but which has proven to be a significant indicator for fraud detection. Rigorous data preprocessing, including the removal of gray samples and ADASYN-based oversampling, ensures a balanced dataset, thereby enhancing both model robustness and performance. Furthermore, this study employed machine learning to develop a detection method represented in the form of a mathematical equation. This equation-based approach enhances the model’s interpretability and user-friendliness, enabling stakeholders to better comprehend fraud dynamics and implement more effective mitigation strategies. It lays the groundwork for transparent, robust, and efficient financial fraud detection systems.

2. Data and methodology

2.1. Data

Financial fraud samples were sourced from the financial annual reports and violation records in the CSMAR database. Table 1 categorizes corporate violations disclosed by the CSRC and other regulatory bodies. To ensure data consistency, only samples from post-2000 were included, as pre-2000 data were incomplete and subject to different policies.

This study focuses on cases up to 2020, excluding the financial industry due to its distinct reporting standards, to ensure consistency and comparability. Seven categories of non-compliant records were carefully selected to represent fraud samples: “Fabricated Profits”, “Fabricated Assets”, “False Records”, “Delayed Disclosure”, “Major Omissions”, “Inaccurate Disclosure”, and “Fraudulent IPO”. Non-fraud samples were rigorously filtered by excluding companies with any non-compliance records, flagged audit reports, or ST/ST* designations. Each year of a company’s behavior was treated as an independent sample, resulting in 5732 fraud samples and 33,198 non-fraud samples. The samples underwent multiple pre-processing steps to ensure data quality and model reliability.

First, gray samples, representing potential undetected fraudulent cases, were excluded using Benford’s Law and the Isolation Forest method to enhance data integrity. Missing values were addressed by removing samples with missing key accounting variables and imputing general missing data using the KNNImputer function. Next, a unified normalization process was applied: robust standardization was used to reduce the influence of outliers, followed by Min-Max scaling to rescale the data into a suitable range

Table 2
Sample distribution of the experiment.

Dataset	Non-fraudulent sample	Fraudulent sample	Artificial sample
Training set	23 653	4299	19 354
Testing set	7884	1433	0

Table 3
Non-financial indicators.

Indicator category	Indicator name
Internal executive information	Proportion of Independent Directors
	Proportion of Management Shareholdings
	Proportion of Supervisor Shareholdings
Shareholding ratio	Proportion of the Largest Shareholder
	Sum of the Proportion of the Top 3 Shareholders
	Sum of the Proportion of the Top 5 Shareholders
	Sum of the Proportion of the Top 10 Shareholders
	Proportion of State-Owned Shareholdings
	Proportion of Other Institutional Holdings in Circulating A Shares
	Proportion of Social Security Fund Holdings in Circulating A Shares
	Proportion of Securities Company Holdings in Circulating A Shares
Management compensation	Proportion of Fund Holdings in Circulating A Shares
	Proportion of the Chairman's Annual Compensation to Total Executive Income
	Proportion of the Top 3 Senior Executives' Total Compensation to Total Executive Income
	Proportion of the Top 3 Directors' Total Compensation to Total Executive Income
	Proportion of Directors' Total Compensation to Total Executive Income
	Proportion of the Board Secretary's Total Compensation to Total Executive Income
	Proportion of the CFO's Total Compensation to Total Executive Income
	Proportion of the Chairman's Compensation to Total Executive Income

for model training. The dataset was subsequently divided using 5-fold cross-validation, and the ADASYN method was employed to balance the training set with a 1:1 ratio of fraudulent to non-fraudulent samples. These steps collectively enhanced the model's performance and robustness. The sample distribution for this study is presented in [Table 2](#).

2.2. Indicator system

As stated in [Zhang et al. \(2020\)](#), identifying fraudulent companies requires selecting indicators that encompass both financial and non-financial risk factors. Therefore we follow the same process.

Financial Indicators

The detection of abnormal financial ratios is a critical approach for identifying financial fraud. Unlike previous methods, our approach emphasizes financial indicators derived from the entire set of data in the balance sheet, income statement, and cash flow statement. All financial ratios generated from these statements are screened and represented as a vector, where each financial ratio corresponds to a specific eigenvalue coordinate. Due to the large volume of financial statement data, the number of generated financial ratios can reach into the thousands. To mitigate the risk of model overfitting, we conduct preliminary screening of these variables. The procedure is as follows:

1. Financial items with missing values exceeding 0.7 are excluded.
2. Duplicate financial items appearing in multiple tables are removed, retaining only one instance.
3. Financial ratios are calculated using pairs of remaining items from the three tables, with the item showing the higher average value serving as the denominator. Ratios are considered missing if the denominator is zero.
4. Ratios with missing values exceeding 0.7 are excluded, except for essential ratios, such as Net Profit Margin, Sales Growth Rate, Return on Assets, Current Ratio, Quick Ratio, Debt-to-Asset Ratio, Accounts Receivable Turnover Ratio, Inventory Turnover Ratio, Operating Cash Flow Ratio, Cash Flow Ratio, and Equity Multiplier.
5. The Point-Biserial Correlation algorithm is then applied to calculate the correlation between each financial ratio and the occurrence of financial fraud. The top 500 financial ratios exhibiting the strongest correlation are selected for further analysis.

Through the above steps, 500 new financial indicators were selected as inputs for the XGB-GP model.

Non-Financial Indicators

The non-financial indicator data used in this study were obtained from the CNRDS and CSMAR databases. As shown in [Table 3](#), a total of 19 non-financial indicators were selected, primarily focusing on information related to internal executives, management compensation, and shareholder ownership ratios. Specifically, internal executive information and management compensation reflect the motivation and ethical standards of the management team ([Qi et al., 2023](#)), while the shareholding ratio provides insights into the company's governance structure and equity distribution ([Li et al., 2024](#)).

The financial and non-financial indicators collectively constitute the comprehensive feature set for this study.

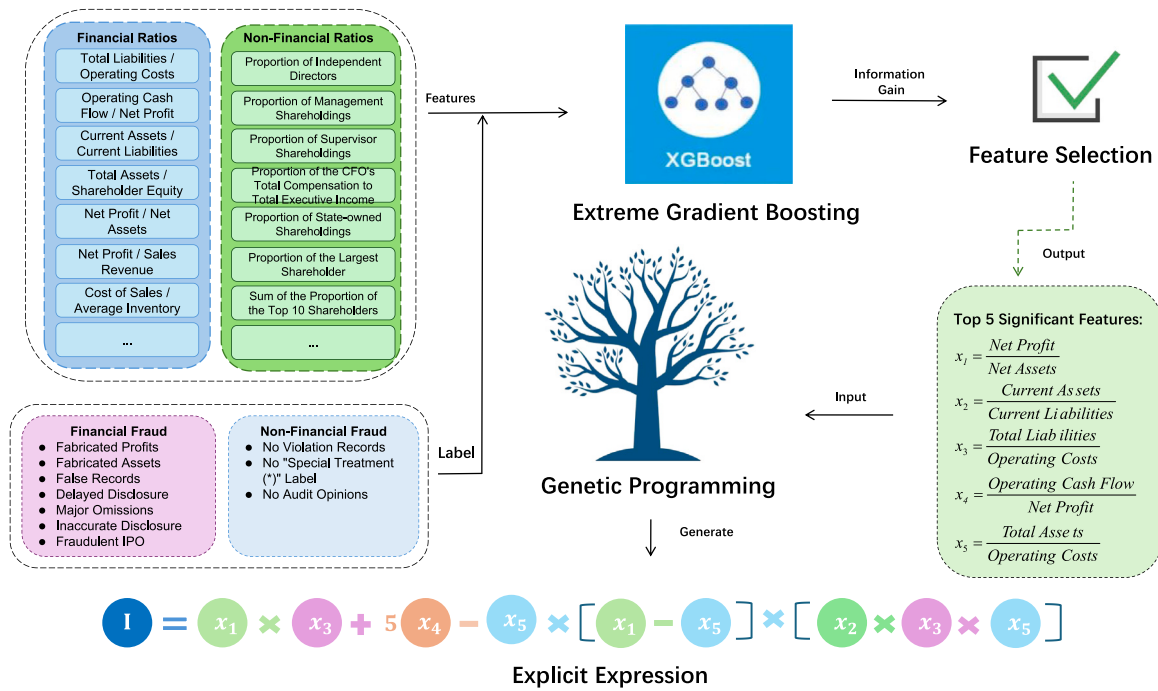


Fig. 1. The proposed XGB-GP framework for financial fraud detection.

2.3. Methodology

The eXtreme Gradient Boosted Genetic Programming (XGB-GP) model constructed in this paper is a hybrid approach combining eXtreme Gradient Boosting (XGBoost) and Genetic Programming (GP) for financial fraud detection. As illustrated in Fig. 1, this methodology employs XGBoost for feature selection, followed by GP to generate a mathematical expression for predicting the target variable.

XGBoost is utilized for feature selection by training a model to predict financial fraud labels and ranking features based on their importance using the gain metric. The top-ranked features are then encoded to form the initial population for GP. Through iterative processes of crossover and mutation, the population evolves to optimize the model's performance (Cui et al., 2013). The best-performing expression from the final population is selected as the fraud detection model, providing a transparent and interpretable solution for identifying financial fraud.

By integrating XGBoost's efficient feature selection with the interpretability of Genetic Programming, the XGB-GP model captures complex nonlinear relationships in a simplified equation format. This hybrid approach offers high accuracy while maintaining user-friendliness for non-technical stakeholders.

3. Empirical results

In this study, the model is based on machine learning, with financial fraud detection treated as a classification problem. The model's performance was evaluated using standard metrics, including Precision, Recall, Accuracy, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC).

3.1. Feature importance

Total information gain is used to measure feature importance. The resulting ranking of feature significance is presented in Fig. 2. Table 4 presents the actual indicators corresponding to the significance ranking labels.

The feature importance graph generated by XGBoost clearly indicates that the top five ratios are significantly more important than the others. Accordingly, this study selects these five ratios as the key features for detecting financial fraud:

1. Return on Equity (ROE), defined as Net Profit divided by Net Assets, ranks first among the top 20 significant factors in this study (Fig. 2). ROE serves as an indicator of a company's profitability, operational efficiency, and debt-paying ability. Persons (1995) analyzed various financial ratios, emphasizing that abnormal fluctuations in ROE are strongly correlated with financial fraud. From a fraud detection perspective, when ROE falls below expectations, management may resort to inflating revenue, concealing expenses, or overstating asset values to artificially improve financial performance. Such actions are often driven by motives such as career advancement, higher compensation, or the desire to sustain stock prices.

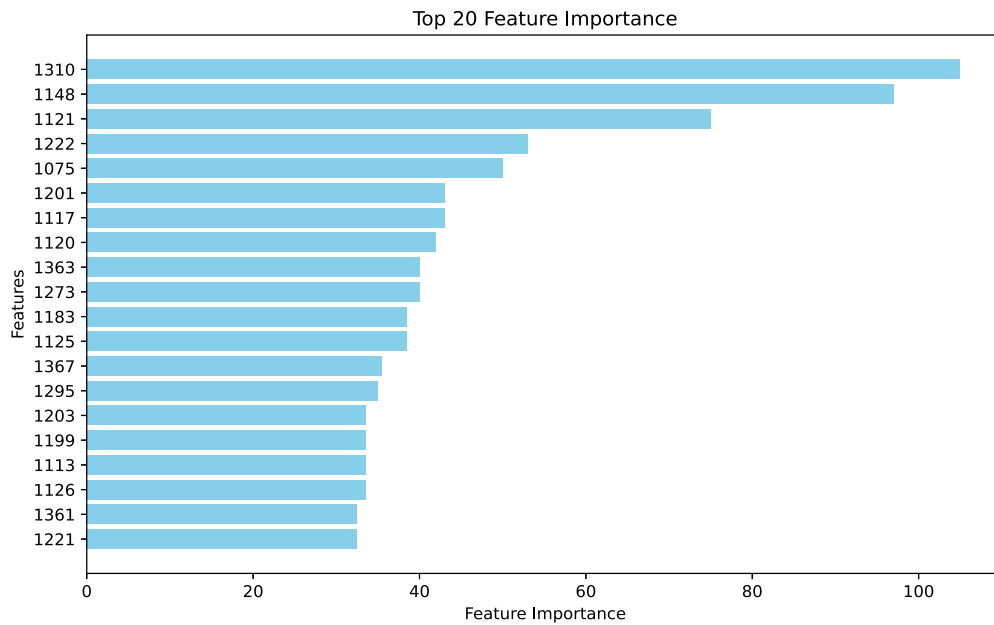


Fig. 2. Important feature selection for financial fraud based on XGBoost.

Table 4

Important feature number correspondence table.

No.	Feature (Ratio)	No.	Feature (Ratio)
1310	Net Profit/Net Assets	1148	Current Assets/Current Liabilities
1121	Total Liabilities/Operating Costs	1222	Operating Cash Flow/Net Profit
1075	Total Assets/Shareholder Equity	1201	Operating Cash Flow/Current Liabilities
1117	Net Profit/Sales Revenue	1120	Operating Profit/Non-operating Income
1363	Other Receivables/Total Assets	1273	Operating Revenue/Inventory
1183	Sum of the Proportion of the Top 10 Shareholders	1125	Inventory/Main Business Costs
1367	Sales Revenue/Total Assets	1295	Net Profit/Total Assets
1203	Operating Profit/Non-operating Expenses	1199	Total Liabilities/Total Assets
1113	Minority Interest/Total Liabilities and Shareholders' Equity	1126	Net Fixed Assets/Net Value Construction in Progress
1361	Long-term Liabilities/Shareholder Equity	1221	Sales Revenue/Average Accounts Receivable

- The Current Ratio (CR), defined as Current Assets divided by Current Liabilities, is a key indicator of a company's short-term solvency. A higher CR suggests robust liquidity, while a lower CR signals weaker solvency. Companies may manipulate this ratio to misrepresent their financial health. [Gullkvist and Jokipii \(2013\)](#) examined the motivations and methods of earnings management, noting that management may adjust financial indicators, such as the Current Ratio, to deceive investors. Inflating the CR can attract investments or loans, while artificially lowering it might conceal liabilities. Abnormal CR values should serve as red flags for auditors and stakeholders, signaling the need for further scrutiny and investigation.
- The ratio of Total Liabilities to Operating Costs, though unconventional, ranks third in significance in this study. Identified by the model for its exceptional predictive power, this new indicator is crucial for assessing a company's financial leverage and debt repayment capacity. A higher ratio indicates greater debt burden and weaker financial health, whereas a lower ratio suggests healthier finances and more efficient resource allocation. From a fraud detection standpoint, companies with lower debt and higher operating costs may appear more attractive to investors, creating an incentive to manipulate this ratio. This indicator provides insights into a company's debt management that may not be captured by traditional ratios like ROA or CR. A disproportionate ratio of debt to operating costs could signal potential risks such as profit manipulation or cash flow distortion. For instance, companies might overstate revenue or underreport expenses to mask financial instability. Monitoring this ratio can aid in the early identification of such fraudulent activities. This finding underscores the importance of advanced detection techniques in addressing the complexities of fraud and mitigating financial risk.
- The ratio of Operating Cash Flow to Net Profit is a key indicator of earnings quality. A higher ratio suggests that profits are well-supported by cash flow, while a consistently low ratio, particularly below 1, may signal profit inflation or difficulties in receivables collection. Companies may manipulate this ratio by employing accounting techniques, such as prematurely recognizing revenue or deferring the recognition of expenses, to artificially inflate profits. Such manipulations, as observed in the 2024 Chemours case, were driven by executive bonuses tied to performance measures like free cash flow ([The Wall Street Journal, 2024](#)).

5. The Equity Multiplier (Total Assets / Shareholders' Equity) measures a company's capital structure and financial leverage, with higher values indicating greater reliance on debt financing and increased financial risk. Van Vlasselaer et al. (2015) examined the relationship between corporate governance and earnings management, identifying the Equity Multiplier as a key indicator of financial leverage, significantly linked to earnings manipulation. Their study found that firms with higher leverage are more likely to face debt repayment pressures, which can motivate management to adopt earnings management strategies to enhance financial performance and conceal the true financial condition.

Non-Financial Indicators

At the same time, we found that non-financial indicators, such as the shareholding ratio of the top ten shareholders (which ranks 11th among the significant factors), have some impact on detecting financial fraud, but their significance is relatively less pronounced. Consequently, in our approach, which prioritizes ease of use, these indicators are excluded from the GP process. However, incorporating such indicators may still offer value in traditional machine learning models that focus on improving prediction accuracy.

3.2. Model evolution using Genetic Programming

3.2.1. Explicit expression for financial fraud detection

Through parameter tuning and model evolution, the GP algorithm constructs an explicit financial fraud detection model, represented as the following expression:

$$I = x_1 \times x_3 + 5 \times x_4 - x_5 \times (x_1 - x_5) \times (x_2 \times x_3 \times x_5 - 1) \quad (1)$$

where x_1, x_2, x_3, x_4, x_5 are the top five features identified by XGBoost (see Fig. 2):

$x_1 = \text{NetProfit} / \text{NetAssets}$

$x_2 = \text{CurrentAssets} / \text{CurrentLiabilities}$

$x_3 = \text{TotalLiabilities} / \text{OperatingCosts}$

$x_4 = \text{OperatingCashFlow} / \text{NetProfit}$

$x_5 = \text{TotalAssets} / \text{ShareholderEquity}$

Since there are two possible outcomes for the fraud detection task (positive/negative), it is necessary to define a boundary for I in Eq. (1). Initial tests reveal that a smaller threshold value generally yields better results. In this study, a threshold of 1 is chosen. Consequently, if I is greater than or equal to 1, it suggests an anomaly in the financial data, which may be indicative of potential fraud.

3.2.2. Discussion

This section aims to interpret and analyze the equation evolved by the XGB-GP model from an economic perspective. The equation consists of three components, each of which will be discussed separately.

1. $x_1 \times x_3$: This term represents the product of return on equity (x_1) and the liabilities-to-operating cost ratio (x_3). A high value of $x_1 \times x_3$ may suggest that a company appears to exhibit strong profitability despite carrying a substantial amount of debt. This combination is commonly observed in fraudulent cases, where companies manipulate financial data by inflating profits or using excessive debt to conceal underlying financial weaknesses.
2. $5 \times x_4$: This term highlights the significance of the operating cash flow to net profit ratio (x_4). A low value of x_4 signals poor earnings quality, suggesting that reported profits may be the result of inflated earnings or non-recurring revenue streams. Such a situation can serve as a warning sign of potential financial manipulation.
3. $x_5 \times (x_1 - x_5) \times (x_2 \times x_3 \times x_5 - 1)$: This term is more complex and requires a breakdown into several components. For the expression $(x_1 - x_5)$, if the return on equity (x_1) exceeds the equity multiplier (x_5), it may indicate that the company is showing high profitability relative to its asset base, a common sign of fraudulent activity. The term $(x_2 \times x_3 \times x_5 - 1)$ incorporates the current ratio (x_2), the liabilities-to-operating cost ratio (x_3), and the equity multiplier (x_5). A high value for this expression may suggest that the company is overly reliant on debt. In fraudulent scenarios, companies might inflate current assets or underreport liabilities to present a more favorable financial position, further distorting the true financial health.

Fraudulent companies often manipulate financial indicators by inflating profits (affecting x_1), overstating assets (affecting x_2 and x_5), or concealing liabilities (affecting x_3) to present a misleadingly favorable financial picture. In the equation, high values of $x_1 \times x_3$ and low values of x_4 may indicate potential fraud. The negative term $-x_5 \times (x_1 - x_5) \times (x_2 \times x_3 \times x_5 - 1)$ highlights discrepancies between profitability, asset-liability ratios, and cash flow quality, which can signal fraudulent activity. This equation aims to detect such fraud by identifying abnormalities in the relationships between total assets, liabilities, the current ratio, and cash flow. Abnormal values in the combination of x_1, x_2, x_3, x_4 , and x_5 may indicate inflated profits, overstated assets, or concealed liabilities. By integrating key financial indicators, the equation helps uncover financial inconsistencies, particularly between profitability, liabilities, and cash flow, making it a powerful tool for identifying potential financial fraud risks.

Beneish (1999) introduced the renowned M-Score model, which detects corporate earnings manipulation by analyzing key financial ratios. These ratios not only assess a company's profitability but also evaluate its liquidity, leverage, and financial stability. The indicators in the equation proposed in this study, such as the net profit to net assets ratio, current ratio, and total assets to equity ratio, closely align with the core principles of the Beneish model. Both approaches aim to identify potential financial fraud

Table 5
Comparison of experimental results for five different models.

Metric	XGB-GP	M-Score	Logistic	SVM	XGBoost	RNN
Accuracy	0.901	0.546	0.668	0.793	0.918	0.813
Precision	0.839	0.325	0.713	0.734	0.867	0.760
Recall	0.884	0.563	0.741	0.841	0.895	0.827
F1 Score	0.828	0.412	0.726	0.783	0.880	0.792
AUC	0.861	0.444	0.727	0.788	0.881	0.794

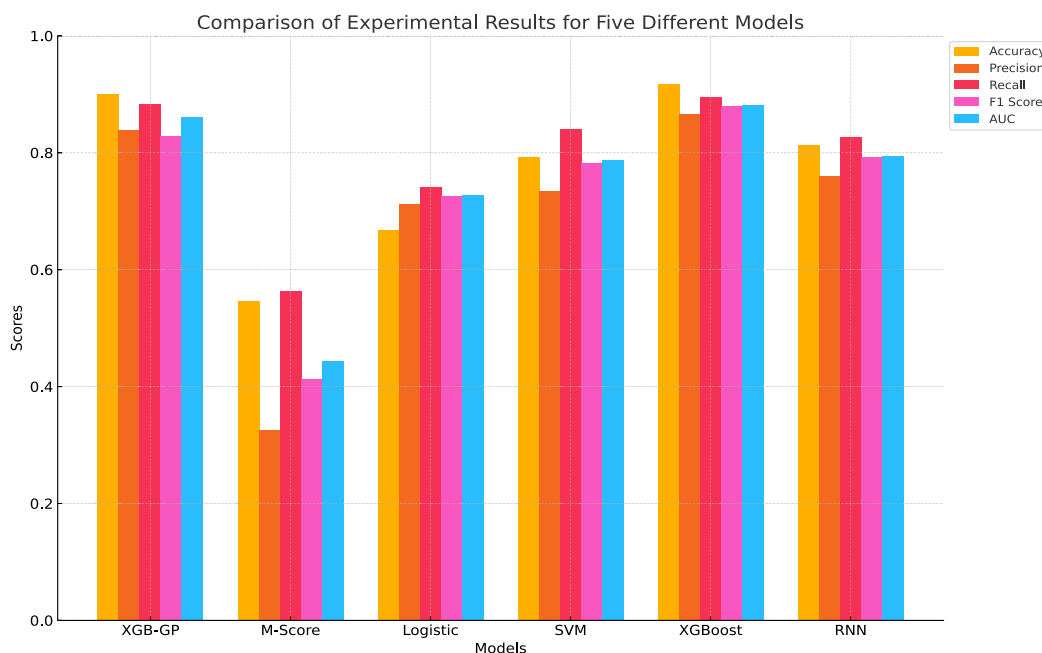


Fig. 3. Comparison of experimental results.

by quantifying financial anomalies. This alignment demonstrates that the equation proposed in this study is grounded in robust theoretical and empirical foundations, sharing a similar logical framework and practical significance with established detection methods in the literature. By synthesizing several key financial indicators, the detection equation evolved from the XGB-GP model effectively reveals financial inconsistencies behind fraud, particularly discrepancies between profitability, liabilities, and cash flow, and can serve as a powerful tool for detecting potential financial fraud risks.

3.3. Comparative experiments

In this study, the M-score, Logistic Regression, SVM, XGBoost, and RNN models were selected as comparison models. Among them, the M-score and Logistic Regression models represent traditional fraud detection methods based on statistical approaches, while SVM, XGBoost, and RNN models represent machine learning techniques. The SVM was chosen for its effectiveness in handling high-dimensional feature spaces. XGBoost, a robust gradient boosting algorithm, is particularly adept at capturing complex nonlinear relationships and serves as the primary model for feature selection in the proposed hybrid framework, making it a key comparison model. Given the temporal nature of financial data, RNNs were included for their ability to capture time-dependent patterns.

In the comparative study, aside from the fixed ratios in the M-score model, the Logistic Regression, SVM, and RNN models all utilized the necessary financial ratios outlined in Section 2.2. In contrast, the XGBoost model employed a feature set comprising the top 500 financial ratios selected using the Point-Biserial Correlation method, along with 16 non-financial ratios from Section 2.2. The XGB-GP model, on the other hand, leveraged five key features selected in Section 3.1. This approach of using a smaller, more refined feature set in the XGB-GP model aims to enhance detection accuracy while preserving interpretability and ease of use. The performance of each model is presented in Table 5 and Fig. 3.

From the perspective of evaluation metrics, XGBoost outperforms all other models, further confirming its effectiveness in fraud detection and underscoring its robust feature selection capability. The XGB-GP model follows closely behind XGBoost, demonstrating high precision and recall, thus validating its strong performance in detecting fraud. Other machine learning models, such as RNN and SVM, show some effectiveness, but do not achieve the same level of overall performance as XGB-GP and XGBoost. In contrast, traditional methods, such as the M-Score and Logistic regression models, perform relatively poorly. This suggests that traditional

approaches struggle to capture the complex nonlinear relationships between features and labels, rendering them less effective in identifying fraudulent activities.

The advantages of the XGB-GP model are evident not only in its superior evaluation metrics but also in its simplicity and transparency. Unlike the XGBoost model, which relies on a large number of features and operates as a black box, the XGB-GP model achieves high performance with only five features and a transparent equation. This makes the XGB-GP model both user-friendly and efficient. Moreover, it presents results through explicit equations, providing detailed economic interpretations of both individual features and the overall equation. This enhances the model's transparency and interpretability, thereby increasing its credibility.

The hybrid model developed in this study automatically selects features using machine learning techniques, minimizing manual intervention and making it highly adaptable to the era of big data. The model is capable of continuously updating its indicator and equations to respond to evolving fraud tactics, enabling it to detect increasingly sophisticated fraudulent activities. Additionally, an effective data processing strategy bolsters the model's training efficiency. Techniques such as gray sample exclusion and data augmentation improve the model's ability to learn fraud characteristics and strengthen its detection capabilities.

By combining high precision, transparency, interpretability, ease of use, and efficiency, the XGB-GP model presents a comprehensive and robust solution for financial fraud detection, outperforming other models.

4. Research implication

This study introduces the XGB-GP model, which demonstrates superior accuracy, precision, and interpretability in financial fraud detection. Unlike traditional “black box” models, our approach leverages XGBoost to identify key characteristic factors and construct a transparent mathematical equation using genetic programming. This makes the model more accessible, allowing users to easily understand and apply it. Compared to conventional algorithms and more complex machine learning models, our model offers distinct advantages in financial fraud detection. It is not only user-friendly for non-technical users but also provides a clear interpretative framework, enabling companies and regulatory bodies to quickly detect and respond to fraudulent activities.

The feature significance analysis in our study provides valuable insights. The “Total Debt/Operating Costs” ratio, ranked third in importance, offers crucial information about a company's financial leverage and debt repayment capacity. This finding highlights the potential value of previously underutilized financial ratios in detecting fraud. Furthermore, it underscores the limitations of relying solely on traditional financial ratios when addressing increasingly sophisticated fraudulent activities.

In a real-time financial fraud detection system, the XGB-GP model supports incremental training, allowing dynamic adjustments to key fraud indicators and the corresponding equations based on incoming data. This allows the model to update its weights and tree structure as the dataset grows, eliminating the need for retraining from scratch. Since the features input into the XGB-GP model are selected using the point-biserial correlation method, interpretability of the updated features is not a concern. The initial statistical filtering ensures that each feature possesses economic significance, thereby guaranteeing the interpretability of the resulting equations. Unlike traditional statistical models, our machine learning-based feature selection and equation construction methods enable the model to efficiently and accurately self-update in response to evolving data. This adaptability allows the model to thrive in big data environments and adjust to changing fraud tactics.

5. Conclusion

Financial fraud and financial risk are closely intertwined, mutually influencing each other and contributing to the risk factors of the financial market. An effective and user-friendly financial fraud detection model can enhance auditing and supervision, improve investors' ability to identify fraudulent activities, and significantly reduce the occurrence of financial fraud. In this work, we introduce the XGB-GP model, which combines XGBoost for feature selection and GP for fraud detection. Our model's user-friendly design, efficiency, and adaptability make it highly suitable for practical applications. We also emphasize the importance of thorough data collection and minimize the impact of false samples on the model. The validity of the model has been confirmed through empirical results. During the research, we also identify a new financial ratio with significant relevance to fraud detection. This finding highlights the complexity of detecting financial fraud and underscore that traditional financial data alone is no longer sufficient to fully uncover fraud risks. Looking ahead, fraud detection will increasingly rely on the integration of multi-dimensional, multi-source data, allowing for the extraction of more comprehensive features from diverse information sources.

CRedit authorship contribution statement

Wenjuan Li: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xinghua Liu:** Validation, Supervision, Software, Resources, Methodology, Formal analysis, Conceptualization. **Junqi Su:** Validation, Methodology, Investigation, Data curation. **Tianxiang Cui:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

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Data availability

Data will be made available on request.

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